MEASUREMENT SENSITIVE
O-I-1279B
June 30, 1993
SUPERSEDING
O-I-1279A(SH)
November 24, 1980
(See 6.4)

FEDERAL SPECIFICATION

ION EXCHANGE RESINS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for use of all Federal Agencies.

1. SCOPE

1.1 Scope. This specification covers general purpose synthetic styrene-divinylbenzene organic anion and cation exchange resins that demineralize or treat water by replacing negative (nonmetal) ions with hydroxide (OH^-) ions or positive (metal) ions with hydrogen (H^+) ions.

1.2 Classification. Ion exchange resins shall be of the following classes, as specified (see 6.2).

Class 1 - Sulfonated cross-linked styrene-divinylbenzene cation exchange resin in the hydrogen (H^+) form.
Class 2 - Trimethylamine type styrene-divinylbenzene quaternary anion exchange resin in the hydroxide (OH^-) form.
Class 3 - Mixed bed. A chemically equivalent mixture of hydrogen form cation resin and hydroxide form anion resin.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY
MS27684 - Drum, Metal Shipping and Storage - 3 to 12 Gallons.

STANDARDS

FEDERAL
FED-STD-123 - Marking for Shipment (Civil Agencies).
FED-STD-313 - Material Safety Data, Transportation Data and Disposal Data for Hazardous Mateials Furnished to Government Activities.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, ATTN: SEA 03Q42, Naval Sea Systems Command, 2531 Jefferson Davis Hwy, Arlington, VA 22242-5160 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6810
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MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized Unit Loads for 40" x 48" Pallets.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 NonGovernment publications. The following documents form a part of this specification to the extent specified herein (see 6.2). Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 1193 - Specification for Reagent Water.
D 2187-82 - Physical and Chemical Properties of Particulate Ion-Exchange Resins.
D 2687 - Sampling Particulate Ion-Exchange Materials.
D 3375 - Column Capacity of Particulate Mixed Bed Ion-Exchange Materials.
D 3951 - Standard Practice for Commercial Packaging.
D 4456 - Physical and Chemical Properties of Powdered Ion-Exchange Resins.
D 4548 - Anion-Cation Balance of Mixed Bed Ion-Exchange Resins.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

STANDARD METHODS FOR THE EXAMINATION OF WATER AND WASTEWATER (17TH EDITION)

Method Fe 3500 D - Iron by Orthophenanthroline.
Method Cu 3500 D - Copper by Neocuproine. SP
Method Na 3500 D - Flame Emission Photometric Method.
Method 4110 B - Ion Chromatography with Chemical Suppression of Eluant Conductivity.

(Application for copies should be addressed to the American Public Health Association, 1015 15th Street, N.W., Washington, DC 20005.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

3. REQUIREMENTS

3.1 Class 1 - Hydrogen-form cation resins. Class 1 resins shall be unused, fully hydrated, styrene-divinylbenzene, sulfonic acid resins in the hydrogen form meeting the following requirements of table I.

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FIGURE 1. Style CWR11 capacitors

TABLE I. Class 1 resin requirements.

Characteristic	Requirements	Test
Solids, percent as received	48.0 ± 3.0	4.4.1
Particle size, percent by volume Retained on 16 mesh Passing through 40 mesh	 2.0 maximum 2.0 maximum	 4.4.2 4.4.2
Available cationic salt-splitting capacity, milliequivalents per gram dry	4.8 minimum	4.4.3
Non-permissible impurities, milligrams per kilogram as received Total ash Iron as Fe Copper as Cu Water extractable	 200 maximum1/ 50 maximum 100 maximum 250 maximum	 4.4.5.1 4.4.5.1.2 4.4.5.1.3 4.4.5.2

1/ The determination of copper and iron may be omitted if the total ash is not greater than 60.

3.2 Class 2 - Hydroxide-form anion resins. Class 2 resins shall be unused, fully hydrated, styrene-divinylbenzene, trimethylamine quaternary, strong base resins in the hydroxide form meeting the following requirements of table II.

TABLE II. Class 2 resin requirements.

Characteristic	Requirements	Test
Solids, percent as received	41.0 ± 3.0	4.4.1
Particle size, percent by volume Retained on 16 mesh Passing through 40 mesh	 2.0 maximum 2.0 maximum	 4.4.2 4.4.2
Total anion capacity, milliequivalents per gram as received Hydroxide, percent of sites Carbonate, percent of sites Chloride, percent of sites Sulfate, percent of sites	1.7 minimum 90.0 minimum 8.0 maximum 2.0 maximum 2.0 maximum	4.4.4.4 4.4.4.1 4.4.4.1 4.4.4.2 4.4.4.3
Non-permissible impurities, milligrams per kilogram as received Total ash Sodium as Na	 200 maximum1/ 100 maximum	 4.4.5.1 4.4.5.1.1

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1/ The determination of sodium may be omitted if the total ash is not greater than 100.

3.3 Class 3 - Mixed bed resins. Class 3 resins shall be a fully hydrated mixture of a sulfonated styrene-divinylbenzene cation resin in the hydrogen form and a styrene-divinylbenzene trimethylamine quaternary anion resin in the hydroxide form mixed at a ratio of one hydrogen site to one hydroxide site (one equivalent H^+ to one equivalent OH^-). The mixture shall conform to the requirements of table III.

TABLE III. Class 3 resin requirements.

Characteristic	Requirements	Test
Solids, percent as received	43.0 ± 3.0	4.4.1
Particle size, percent by volume Retained on 16 mesh Passing through 40 mesh	 2.0 maximum 2.0 maximum	 4.4.2 4.4.2
Anion-cation balance, milliequivalents per milliequivalent as received	1.1 ± 0.05	4.4.6
Cation exchange capacity, milliequivalents per gram as received	0.95 minimum	4.4.7
Anion exchange capacity, milliequivalents per gram as received	1.0 minimum	4.4.8
Column capacity, milliequivalents per gram as received	0.75 minimum	4.4.9
Anion impurities, Chloride, percent of anion sites Sulfate, percent of anion sites	 2.0 maximum 2.0 maximum	 4.4.4.2 4.4.4.3
Non-permissible impurities, milligrams per kilogram as received Total ash Sodium as Na Iron as Fe Copper as Cu	 200 maximum1/ 100 maximum 100 maximum 50 maximum	 4.4.5.1 4.4.5.1.1 4.4.5.1.2 4.4.5.1.3

1/ The determination of the specific components may be eliminated if the total ash is not greater than 60.

3.4 Unit content. The unit content shall be one cubic foot of resin.

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3.5 Material Safety Data Sheet. The contracting activity shall be provided a material safety data sheet (MSDS) at the time of contract award. The MSDS shall be provided in accordance with the requirements of FED-STD-313. The MSDS shall be included with each shipment of the material covered by this specification (see 5.6 and 6.3).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure suppliles and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet the requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's cverall inspection system or quality program. The absence of any inspection requirement in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Quality conformance inspection. Quality conformance inspection shall consist of the examination of 4.2.2, the tests of 4.4 and the inspection of 4.5.

4.2.1 Lot. For the purposes of examination, testing and inspection, a lot shall consist of all resin from one production batch of a single class offered for delivery at one time.

4.2.2 Examination of filled containers. As a minimum, the contractor shall randomly select a sample quantity of filled containers in accordance with table IV and examine them for defects of the container and the closure, for evidence of leakage, and for unsatisfactory markings. If one or more defects are found in any sample, the entire lot represented by the sample shall be rejected. If a lot is rejected, the contractor has the option of screening 100 percent of the lot for the defective characteristic(s) or providing a new lot which shall be examined in accordance with the sampling plan contained herein. The contractor shall maintain for a period of three years after contract completion all records of inspections, tests, and any resulting rejections.

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REQUIREMENTS:

TABLE IV. Sampling for examination of filled containers.

Lot size	Sample size 1/2/
2 - 50	5
51 - 90	7
91 - 150	11
151 - 280	13
281 - 500	16
501 - 1200	19
1201 - 3200	23
3201 - 10000	29
10001 - 35000	35
35001 - Over	40

1/ All defective items shall be replaced with acceptable items prior to lot acceptance.
2/ Inspect sample size until reject criteria is reached.

Each sample filled container shall be weighed to determine conformance with the unit content requirement of 3.4. Determine the net weight of wet resin in pounds by subtracting the average tare weight of the empty container from the filled weight. Convert the net weight of resin to cubic feet as follows:

$$\text{Cubic feet} = \frac{\text{Net weight of wet resin, pounds}}{\text{Class factor}}$$

Where:
Class 1 resin class factor = 50
Class 2 resin class factor = 43
Class 3 resin class factor = 43

4.2.3 Sampling for tests. Sampling of a lot shall be in accordance with ASTM D 2687 method A and the procedures of section 8 for moist materials. Two samples shall be selected and packaged; one for tests and one for the contractor. Each sample shall contain not less than one liter of resin for classes 1 and 2 and not less than two liters of resin for class 3. The quartering procedure of section 9 shall be omitted but each sample shall contain material from every sampled package. When package markings allow, sampling of class 3 resin shall include a 500 milliliter (mL) sample from each of the first and the last packages of the lot. These samples shall not be composited but shall be retained in separate sealed containers, marked with the number of the drums from which they were sampled and used for the anion-cation balance of 4.4.6.

4.2.3.1 When Government tests are required, samples shall be forwarded to a laboratory satisfactory to the acquisition activity with a copy of the contractors data (see 6.2).

4.3 Retests and rejection. Failure of the material to conform to any one of the tests of this specification during inspection shall be cause for rejection of the entire lot represented by the sample. If, in the opinion of the contractor, the failure was due to a faulty test method, the tests may be repeated on the contractors retained sample. If that sample also fails, the lot represented by the sample shall be rejected.

4.4 Test procedures.

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4.4.1 Percent solids (classes 1, 2, and 3). The percent solids content shall be determined in accordance with ASTM D 4456 method B (see 4.4.3).

4.4.2 Particle size (classes 1, 2, and 3). The particle size distribution shall be determined in accordance with ASTM D 2187-82 method D or by an optical sizing method that sizes and counts particles passing through a collimated light beam in an aqueous stream. In the event of controversy, the optical sizing method shall be the referee method. Optical sizing instruments for sizing ion exchange resins are available from HIAC/ROYCO, Division of Pacific Scientific, 141 Jefferson Drive, Menlo Park, CA 94025.

The optical sizing instrument shall have a minimum of six channels but the use of a 32 channel instrument is recommended. The range of the channels shall be set from 1400 micrometers to 150 micrometers. An E2500 sensor is suitable for particle size measurement. The optical sizing instrument shall be calibrated with a minimum of three certified samples of known particle size ranging from 1000 to 300 micrometers. Suitable calibration standards are available from Duke Scientific Corporation, P.O. Box 50005, Palo Alto, CA 94303.

Three samples, each weighing not less than 1.5 grams (g), of the as received resin (wet) shall be measured. Each sample shall be fed to the sensor at a concentration not greater than 4 particles per milliliter of water while stirring at a rate to keep the particles in suspension without sucking air into the system. The feed system shall be constructed so as to feed the entire sample.

When the instrument automatically calculates the volume percentages from the number of particles per channel, the values for the volume percent retained on 16 mesh (1190 micrometers) and through 40 mesh (420 micrometers) for the individual portions counted shall be averaged and the average reported.

When the volume percentages are not automatically calculated, the volume of particles in a channel shall be obtained by multiplying the number of particles counted in the channel by the appropriate factor for the channel. Factors to convert to volume for each channel are determined by first calculating the volume factor for the points at which the channels are set. The square root of the product of the volume factors for the two channel limits is then used as the factor to calculate the volume of particles between them. The resin volumes of each channel are added and the volume percentages retained in each channel calculated in accordance with ASTM D 2187-82 method D. When graphing the results, the convention that the volume passing through 20 mesh and retained on 30 mesh is plotted at 30 mesh (600 micrometers) shall be followed. Each sample shall be counted and calculated individually and the results averaged to obtain the values to be reported.

4.4.3 Available cationic salt-splitting capacity (class 1). The available cationic salt-splitting capacity shall be determined in accordance with sections 55.1 through 55.1.3 of ASTM D 2187-82 method G. Calculate the salt-splitting capacity in milliequivalents per gram dry (meq/g dry) from E_R, defined as the average titrations of section 55.1.3, and the percent solids determined in 4.4.1 as follows:

$$\text{Cation salt-splitting capacity, meq/g dry} = \frac{E_R \times N_{NaOH}}{\% \text{ Solids}} \times 100$$

Three samples shall be calculated individually and their average reported.

4.4.4 Anion exchange capacities (class 2 and 3).

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4.4.4.1 Percent of sites in hydroxide and carbonate forms (class 2). The percent of anion sites in the hydroxide and carbonate forms shall be determined in accordance with ASTM D 2187-82 method K (see 4.4.4.2, 4.4.4.3, 4.4.4.4 and 4.4.6). Calculate the percent sites as follows with R, S, C and D as defined in section 90.1:

$$\text{Percent hydroxide sites} = \frac{(2R - S) \times {}^{N}HCl \times 100}{(C - D) \times {}^{N}AgNO_3}$$

$$\text{Percent carbonate sites} = \frac{(S - R) \times {}^{N}HCl \times 200}{(C - D) \times {}^{N}AgNO_3}$$

Three samples shall be calculated individually and their average reported.

4.4.4.2 Percent of sites in chloride form (class 2 and 3). Determine the percent chloride sites in accordance with ASTM D 2187-82 method J or Standard Method 4110 B. The solution generated in section 89.1 of ASTM D 2187-82 method K may be used as well as the values of C and D determined in 4.4.4.1. When using Standard Method 4110 B, analyze for both chloride and sulfate simultaneously (see 4.4.4.3) by diluting 5 mL of the solution generated in section 89.1 to 100 mL. Standards shall be prepared in the same dilution of the sodium nitrate used in the elution. Three samples shall be calculated individually and their average reported.

4.4.4.3 Percent of sites in sulfate form (class 2 and 3). Determine the percent sulfate sites in accordance with ASTM D 2187-82 method L or Standard Method 4110 B. The solution generated in section 89.1 of ASTM D 2187-82 method K may be used as well as the values of C and D determined in 4.4.4.1. A single 200 mL aliquot is used from each sample. When 200 mL aliquots are used, the calculation may be simplified to:

$$\text{Percent sulfate sites} = \frac{428 \times W}{(C - D) \times {}^{N}AgNO_3}$$

When using Standard Method 4110 B, analyze for both chloride and sulfate simultaneously (see 4.4.4.2) by diluting 5 mL of the solution generated in section 89.1 to 100 mL. Standards shall be prepared in the same dilution of the sodium nitrate used in the elution. Three samples shall be calculated individually and their average reported.

4.4.4.4 Total anion capacity (class 2). The total anion capacity in milli-equivalents per gram as received (meq/g wet) shall be determined from the values of C and D obtained in ASTM D 2187-82 method K (see 4.4.4.1) as follows:

$$\text{Total anion capacity, meq/g wet} = \frac{(C - D) \times {}^{N}AgNO_3}{2}$$

Three samples shall be calculated individually and their average reported.

4.4.5 Non-permissible impurities (class 1, 2 and 3).

4.4.5.1 Total ash (class 1, 2 and 3). Weigh approximately 50 g of as received resin into an ignited and weighed (plus or minus 0.0001 g) platinum dish. Dry in an oven at 104 ± 2°C for 2 hours. Heat on a Meker burner at low flame until the resin is thoroughly blackened. Add sulfuric acid if necessary, heating at full flame.

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Continue igniting until the resin beads are burned away. Cool in a desiccator and reweigh. Calculate the total ash in milligrams per kilogram as received (mg/kg wet) as follows:

$$\text{Total ash, mg/kg wet} = \frac{A - C}{B - C} \times 1{,}000{,}000$$

Where:
A = Weight of dish plus ash.
B = Weight of dish plus sample.
C = Weight of dish.

Report the average of two samples.

4.4.5.1.1 Sodium as Na (class 2 and 3). Extract the ash formed in 4.4.5.1 with approximately 50 mL of ASTM D 1193 type I reagent water at 90°C. Filter the extract through Whatman No. 41 filter paper collecting the filtrate in a 100 mL volumetric flask. Dilute to volume. Save the paper for 4.4.5.1.2. Determine sodium in the filtrate by flame emission spectroscopy, Method Na 3500 D, or by ion chromatography, Method 4110 B. Calculate sodium as Na in mg/kg wet as follows:

$$\text{Sodium as Na, mg/kg wet} = \frac{\text{Sodium as Na, milligrams/100 mL}}{\text{Grams of sample used}} \times 1{,}000$$

Report the average of two samples.

4.4.5.1.2 Iron as Fe (class 1 and 3). If sodium was extracted (see 4.4.5.1.1) return the paper with residue to a dish and burn away the paper. Otherwise, fuse the ash residue directly with approximately two grams of potassium pyrosulfate. Dissolve the cooled melt in hydrochloric acid (1 ± 9) and dilute to 100 mL for the determination of iron and copper (see 4.4.5.1.3). Determine iron spectrophotometrically using orthophenanthroline, Method Fe 3500 D. Calculate iron as Fe in mg/kg wet as follows:

$$\text{Iron as Fe, mg/kg wet} = \frac{\text{Iron as Fe, milligrams /100 mL}}{\text{Grams of sample used}} \times 1{,}000$$

Report the average of two samples.

4.4.5.1.3 Copper as Cu (class 1 and 3). Using the solution prepared in 4.4.5.1.2 determine copper by neocuproine, Method Cu 3500 D. Calculate copper as Cu in mg/kg/wet as follows:

$$\text{Copper as Cu, mg/kg wet} = \frac{\text{Copper as Cu, milligrams/100 mL}}{\text{Grams of sample used}} \times 1{,}000$$

Report the average of two samples.

4.4.5.2 Water extractables (class 1). Weigh a 50.0 g sample of the as received resin into a stoppered 250 mL flask and add 200 mL of deionized water. Stopper and heat for 16 hours in a 60 ± 2°C water bath. Filter through a porosity A sintered glass funnel. Evaporate 100 mL of filtrate to dryness at 104 ± 2°C in a weighed platinum dish. A 100 mL blank of the deionized water used shall be similarly evaporated at the same time. Reweigh the dishes. Calculate the residue in mg/kg wet as follows:

$$\text{Residue, mg/kg wet} = (A - B) \times 40$$

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Where:
A = Weight of the sample residue.
B = Weight of the blank residue.

Report the average of two results.

4.4.6 Anion-cation balance (class 3). The anion-cation balance shall be determined in accordance with ASTM D 4548 (see 4.4.7 and 4.4.8). The solution generated in section 81.5 when using ASTM D 2187-82 method K (see 4.4.4.1) is the same as the solution produced in section 8.4 of ASTM D 4548 and may be used in the titration of section 8.5. Also, the values of C and $^{N}AgNO_3$ may be used as F and N_S in the calculation of section 9.1.

4.4.7 As received cation exchange capacity (class 3). The cation capacity in meq/g wet shall be determined using the values from 4.4.6 as follows:

$$\text{Cation capacity, meq/g wet} = \frac{E \times N_B}{2}$$

4.4.8 As received anion exchange capacity (class 3). The anion capacity in meq/g wet shall be determined using the values from 4.4.6 as follows:

$$\text{Anion capacity, meq/g wet} = \frac{F \times N_S}{2}$$

4.4.9 Column capacity (class 3). The column capacity shall be determined in accordance with ASTM D 3375. Omit the sodium silicate from the test water and measure only the conductivity of the effluent. As the capacity is based on the original sample weight (270 ± 1 g), the final backwashing step determining the sample volume, S, may be omitted. Calculate the capacity in meq/g wet as follows:

$$\text{Column capacity, meq/g wet} = \frac{C \times V}{270}$$

Where:
C = The chloride equivalency of the test water (since no silica is used).
V = Volume of test water used in liters.

4.5 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing, and marking for shipment, stowage, and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(Preparation for delivery requirements specified herein apply only for direct Government acquisitions.)

5.1 General requirements. Ion exchange resin shall be furnished in 1-cubic foot quantities as specified (see 3.4) and packaged in double polyethylene bags. Each bag thickness shall be not less than 6 mil and each bag closure shall be by heat sealing.

5.2 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.2.1 Level A. Each unit content bagged as specified in 5.1 shall be packaged in a metal drum conforming to MS27684-7.

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5.2.2 Commercial. Commercial packaging shall be in accordance with ASTM D 3951 (see 6.2).

5.3 Packing. Packing shall be levels A, B, C, or commercial as specified (see 6.2).

5.3.1 Levels A, B and C. Unless otherwise specified (see 6.2) ion exchange resin packaged as specified in 5.2.1 shall not require further packing for shipment.

5.3.2 Commercial. Commercial packing shall be in accordance with ASTM D 3951.

5.4 Palletized unit loads. When specified (see 6.2) ion exchange resin packed as specified in 5.3 shall be palletized in accordance with MIL-STD-147.

5.5 Marking.

5.5.1 Military. In addition to any special marking required (see 6.2) and herein (see 5.5.3), interior packages, exterior shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 for levels A, B and C. Commercial marking shall be in accordance with ASTM D 3951 and shall include applicable bar code markings.

5.5.2 Civil agencies. In addition to any special marking required (see 6.2) and herein (see 5.5.3), marking shall be in accordance with FED-STD-123.

5.5.3 Special marking. Interior packages and exterior shipping container markings shall include the following (see 6.2):

(a) Class of resin.
(b) Resin type.
(c) Batch number.
(d) Shelf life (2 years from date of manufacture).
(e) Date of manufacture.
(f) Net content.
(g) Store resin at temperatures between 32°F and 100°F (class 2 and 3).

5.6 Material safety data sheet. A copy of the MSDS shall be attached to the shipping document for each destination (see 3.5).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The ion exchange resin is intended tc be used to demineralize or treat water by replacing nonmetal (negatively charged) ions with hydroxide ions (OH^-) using class 2 resin or by replacing metal (positively charged) ions with hydrogen ions (H^+) using class 1 resin. Class 3 resin is a chemically equivalent mixture of hydroxide form and hydrogen form resins which replace salt solutions with electrically neutral water molecules. The class 3 resin is suitable for use in naval shipboard (non-nuclear) demineralization units.

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Class required (see 1.2).

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(c) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(d) When Government tests are required (see 4.2.3.1).
(e) Applicable level of preservation and packing (see 5.2 and 5.3).
(f) Commercial packaging requirements (see 5.2.2).
(g) Packing other than specified (see 5.3.1 and 5.3.2).
(h) Commercial packing requirements (see 5.3.2).
(i) Palletization, if required (see 5.4).
(j) Special marking required (see 5.5.1, 5.5.2 and 5.5.3).
(k) Naval shipboard use, if intended (see 6.1).

6.3 Material safety data sheet. Contracting officers must identify those activities requiring copies of MSDSs. Additional required Government information is contained in FED-STD-313. In order to obtain the MSDS, FAR clause 52.223-3 must be in the contract.

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

MILITARY INTERESTS:

Custodians:

Navy - SH

Review Activity:
Army - SM

CIVIL AGENCY COORDINATING ACTIVITY:

GSA-FSS

Preparing activity:
Navy - SH
(Project 6810-1283)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	0-I-1279B	JUNE 30, 1993

3. DOCUMENT TITLE
ION EXCHANGE RESINS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME TECHNICAL POINT OF CONTACT (TOPC): MR. ALBELFIELD (03X22) PLEASE ADDRESS ALL CORRESPONDENCE TO:	b. TELEPHONE (Include Area Code) (1) Commercial: (703) 602-1592 (2) AUTOVON: 8-332-1592
c. ADDRESS (Include Zip Code) COMMANDER, SEA 05Q42 NAVAL SEA SYSTEMS COMMAND 2531 JEFFERSON DAVIS HWY ARLINGTON, VA 22242-5160	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198/290

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